

Proof of Bernoulli's Theorem

Let l denote the sum of the terms in the expansion of $(p + q)^n$ (where $p + q = 1$), from the term containing q^h up to the term containing $q^{h'}$. It is easy to see that

$$l = \frac{1}{2\pi} \int_{-\pi}^{+\pi} [p + qe^{z\sqrt{-1}}]^n e^{-(h+h')\frac{z}{2}\sqrt{-1}} \frac{\sin(h' - h + 1)\frac{z}{2}}{\sin \frac{z}{2}} dz. \quad (1)$$

This expression can be put into the following other form, by setting $h = b - c$, $h' = b + c$:

$$l = \frac{1}{\pi} \int_0^\pi [p^2 + q^2 + 2pq \cos z]^{\frac{n}{2}} \cos(n\psi - bz) \frac{\sin(c + \frac{1}{2})z}{\sin \frac{z}{2}} dz, \quad (2)$$

where

$$\psi = \arctg \frac{q \sin z}{p + q \cos z}.$$

From this one obtains

$$\psi = qz - Qz^3 + \dots, \quad (3)$$

writing, for shortness,

$$Q = \frac{1}{3}q^3 - \frac{1}{2}q^2 + \frac{1}{6}q.$$

Furthermore,

$$\begin{cases} \log(p^2 + q^2 + 2pq \cos z)^{\frac{n}{2}} = -\frac{n}{2}pqz^2 + n \left(\frac{pq}{24} - \frac{p^2q^2}{4} \right) z^4 + \dots, \\ (p^2 + q^2 + 2pq \cos z)^{\frac{n}{2}} = e^{-\frac{n}{2}pqz^2} \left(1 + n \left(\frac{pq}{24} - \frac{p^2q^2}{4} \right) z^4 + \dots \right), \end{cases} \quad (4)$$

$$\frac{1}{\sin \frac{z}{2}} = \frac{2}{z} \left(1 + \frac{z^2}{24} + \dots \right), \quad (5)$$

and therefore

$$l = \frac{2}{\pi} \int dz e^{-\frac{n}{2}pqz^2} \frac{\sin(c + \frac{1}{2})z}{z} \left(1 + \frac{z^2}{24} + nkz^4 + \dots \right) \cos((nq - b)z - nQz^3 + \dots). \quad (6)$$

Let $nq = b$ and $c + \frac{1}{2} = \gamma\sqrt{n}$, $z = \varphi/\sqrt{n}$. Then

$$l = \frac{2}{\pi} \int d\varphi e^{-\frac{pq}{2}\varphi^2} \frac{\sin \gamma\varphi}{\varphi} \left(1 + \frac{\varphi^2}{24n} + k\frac{\varphi^4}{n} + \dots \right) \left(1 - n^2 \frac{Q\varphi^6}{n^3} + \dots \right), \quad (7)$$

$$l = \frac{2}{\pi} \int_0^\infty d\varphi e^{-\frac{pq}{2}\varphi^2} \frac{\sin \gamma\varphi}{\varphi} = \frac{2}{\sqrt{\pi}} \int_0^{\frac{\gamma}{\sqrt{2pq}}} e^{-u^2} du. \quad (8)$$

The probability that, in n simple events, the event B (whose probability is q) lies between

$$qn \pm \gamma\sqrt{n}$$

is therefore

$$\frac{2}{\sqrt{\pi}} \int_0^{\frac{\gamma}{\sqrt{2pq}}} e^{-u^2} du.$$

Equivalently, if $\gamma/\sqrt{2pq} = \beta$, then

$$\frac{2}{\sqrt{\pi}} \int_0^\beta e^{-u^2} du$$

is the probability that Q/n lies between

$$q \pm \frac{\beta\sqrt{2pq}}{\sqrt{n}}.$$

Remarks on the Preceding Memoir

1) Formula (1) can be verified as follows. Rewrite the value of l as

$$l = \sum_{\alpha=0}^n \binom{n}{\alpha} p^{n-\alpha} q^{\alpha} \frac{1}{2\pi} \int_{-\pi}^{+\pi} e^{\alpha z i} e^{-(h+h') \frac{z i}{2}} \frac{\sin(h' - h + 1) \frac{z}{2}}{\sin \frac{z}{2}} dz.$$

Set

$$J_{\alpha} = \frac{1}{2\pi} \int_{-\pi}^{+\pi} e^{\alpha z i - (h+h') \frac{z i}{2}} \frac{\sin(h' - h + 1) \frac{z}{2}}{\sin \frac{z}{2}} dz.$$

With $e^{z i} = u$ this becomes

$$J_{\alpha} = \frac{1}{2\pi i} \int u^{\alpha-1-h'} (1 + u + u^2 + \dots + u^{h'-h}) du,$$

where the integration is over the circumference of the unit circle surrounding $u = 0$.
Hence

$$J_{\alpha} = 0 \quad (\alpha < h \text{ or } \alpha > h'), \quad J_{\alpha} = 1 \quad (h \leq \alpha \leq h').$$

Thus indeed

$$l = \binom{n}{h} p^{n-h} q^h + \binom{n}{h+1} p^{n-h-1} q^{h+1} + \dots + \binom{n}{h'} p^{n-h'} q^{h'}.$$

2) Substituting $h = b - c$, $h' = b + c$ in (1) gives formula (2), after writing P for the modulus of $p + qe^{z i}$ and ψ for its argument.

3) For (6), expansion of the two factors gives

$$l = \frac{2}{\pi} \int_0^{\pi} dz e^{-\frac{n}{2} p q z^2} \frac{\sin(c + \frac{1}{2})z}{z} \left[1 + n \left(\frac{p q}{24} - \frac{p^2 q^2}{4} \right) z^4 + \dots \right] \\ \times \left[1 + \frac{z^2}{24} + \frac{7z^4}{5760} + \dots \right] \cos((nq - b)z - nQz^3 + \dots),$$

so that in order to obtain the formula stated in the text one must put

$$nk = n \left(\frac{p q}{24} - \frac{p^2 q^2}{4} \right) + \frac{7}{5760}.$$

4) In (7), $Q^2/2$ must be put in place of Q ; that is, the final factor must read

$$\left(1 - \frac{n^2 Q^2 \varphi^6}{2n^3} + \dots \right).$$

5) In (8) it is first assumed that n is very large. The second equality follows from the known formula

$$\int_0^{\infty} e^{-x^2} \cos 2\alpha x dx = \frac{\sqrt{\pi}}{2} e^{-\alpha^2},$$

whence

$$\frac{d}{d\alpha} \int_0^{\infty} e^{-x^2} \frac{\sin 2\alpha x}{x} dx = \sqrt{\pi} e^{-\alpha^2}.$$

Since the integral vanishes for $\alpha = 0$, one obtains

$$\int_0^{\infty} e^{-x^2} \frac{\sin 2\alpha x}{x} dx = \sqrt{\pi} \int_0^{\alpha} e^{-u^2} du.$$

Putting $x = \varphi\sqrt{pq/2}$ and $\alpha = \gamma/\sqrt{2pq}$ gives (8).

6) In the third-to-last line on p. 355 the letter Q occurs in a sense different from that used in equations (3), (6), and (7): it is the number of occurrences of the event B among the n trials. Thus l is the probability that

$$h = b - c \leq Q \leq b + c = h',$$

or, after $b = nq$ and $c + 1/2 = \gamma\sqrt{n}$ have been chosen, that Q lies between the limits

$$qn \pm (\gamma\sqrt{n} - \frac{1}{2}),$$

or, rounded,

$$qn \pm \gamma\sqrt{n}.$$

For this reason, in the penultimate line of the text $\sqrt{n}$ belongs not in the numerator but in the denominator: l is the probability that the ratio Q/n lies between

$$q \pm \frac{\gamma}{\sqrt{n}} = q \pm \frac{\beta\sqrt{2pq}}{\sqrt{n}}.$$